

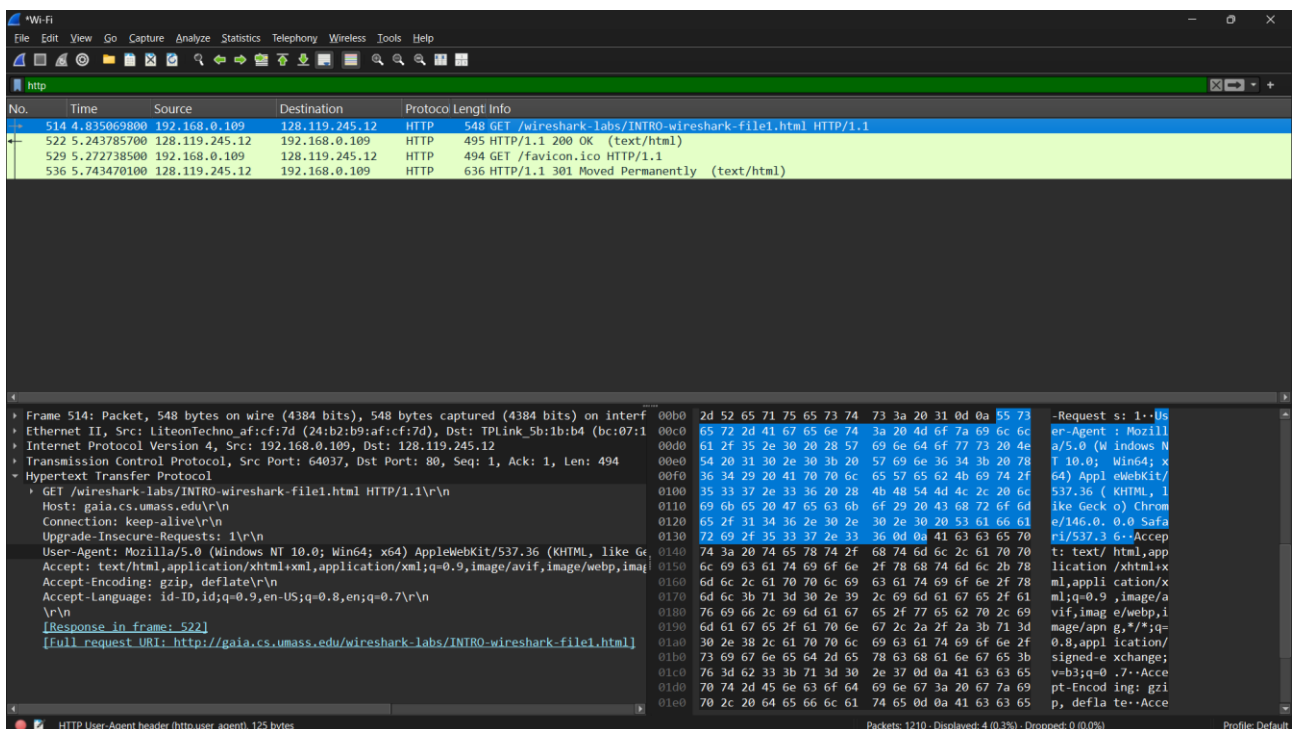
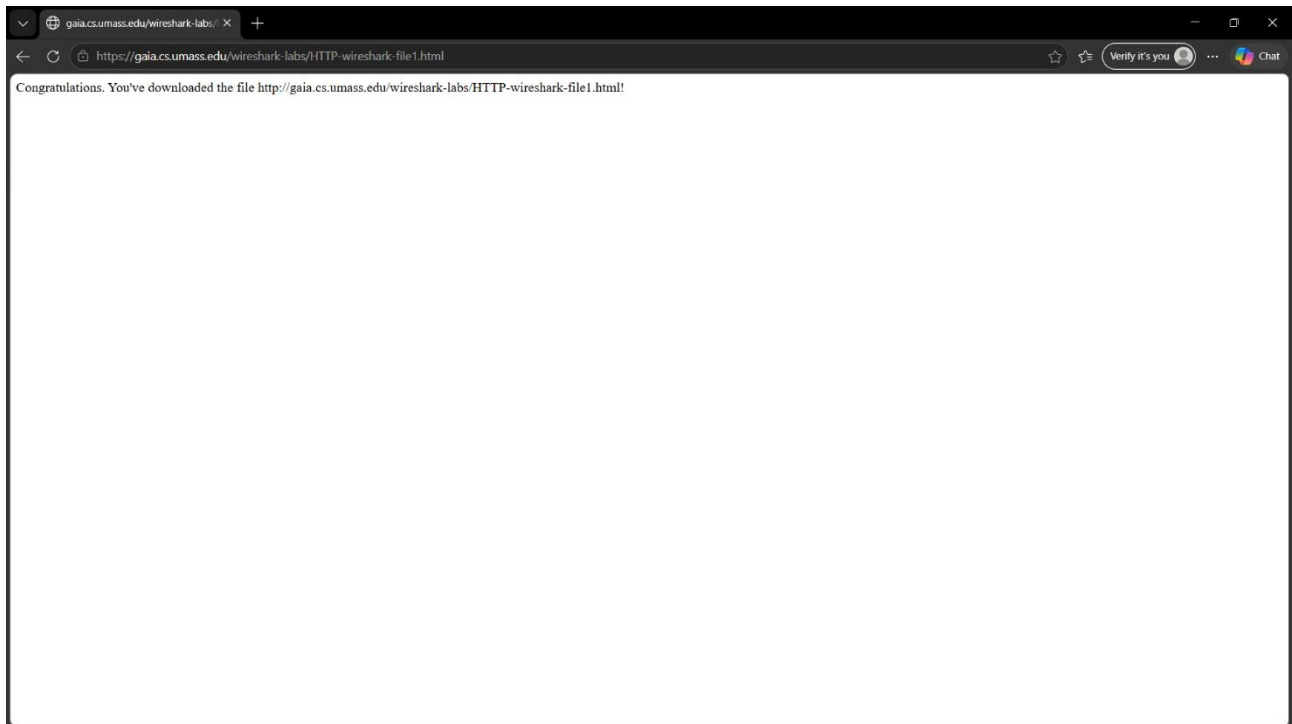
Laporan Praktikum Jaringan Komputer WEEK 3

HTTP

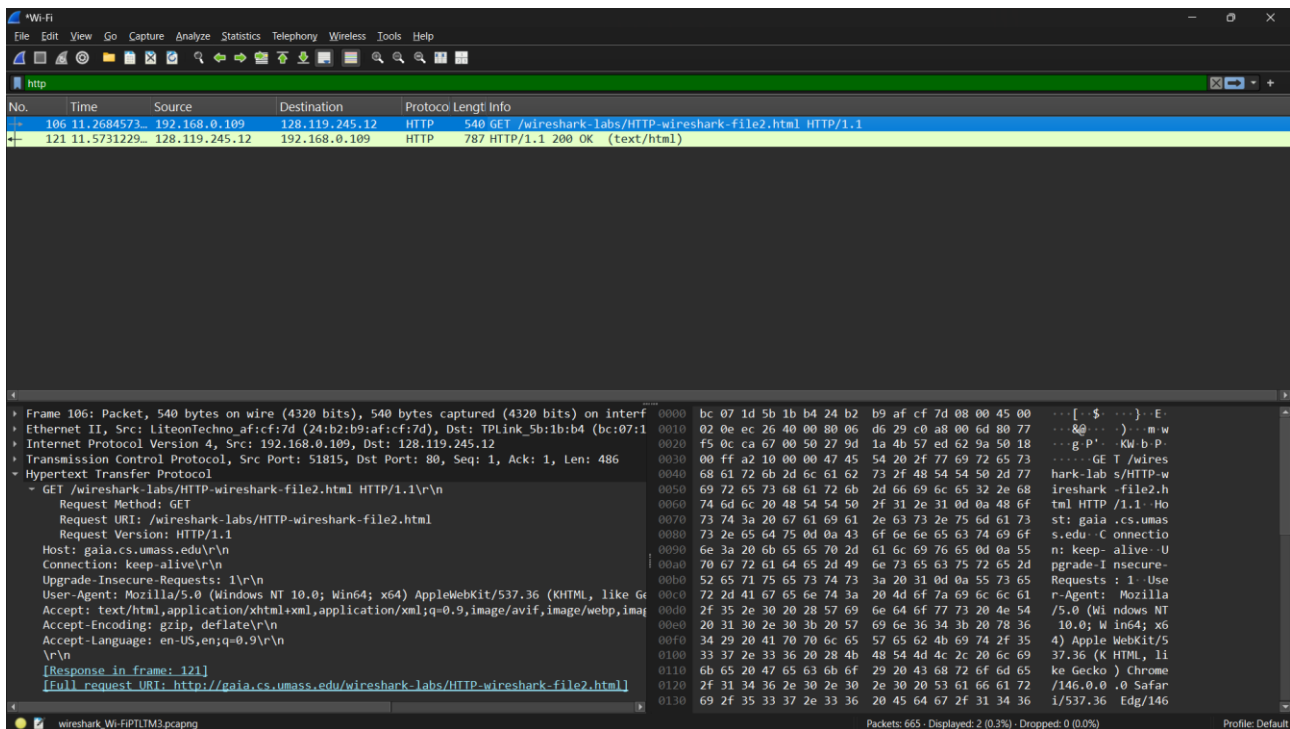
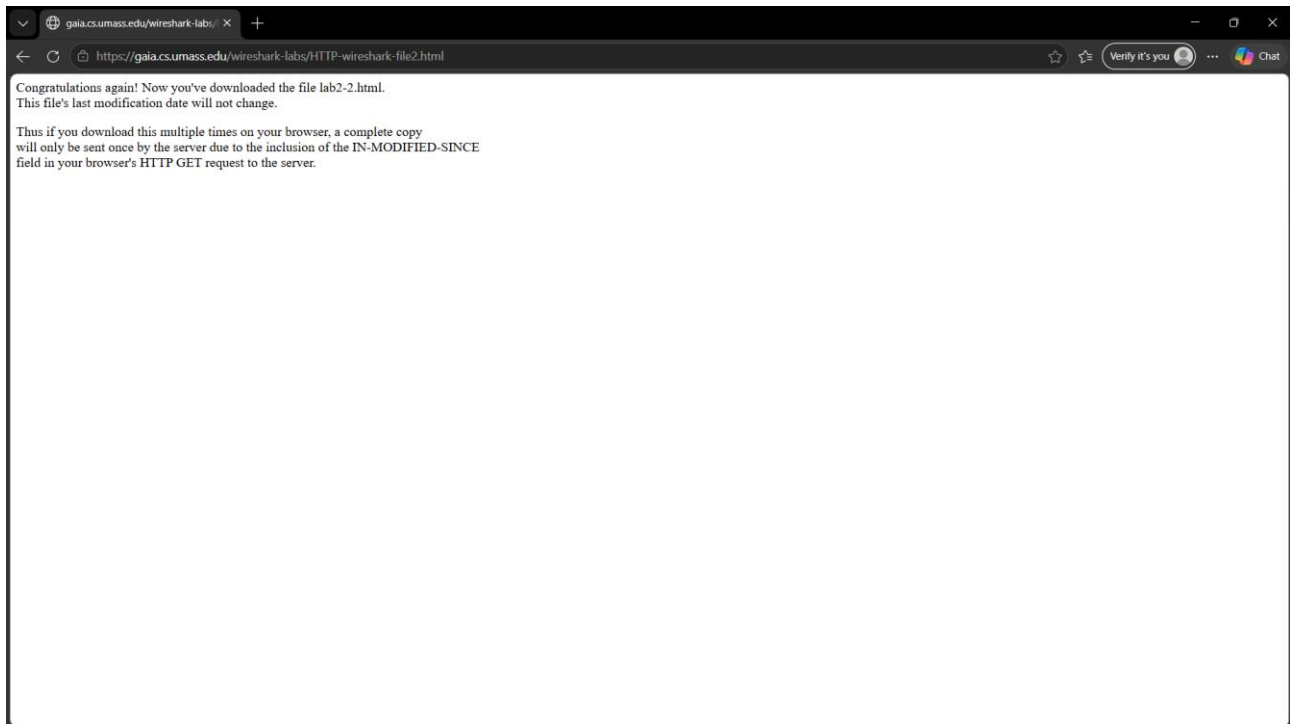
Tujuan Praktikum :

1. Mahasiswa dapat menginvestigasi cara kerja protokol HTTP menggunakan Wireshark.

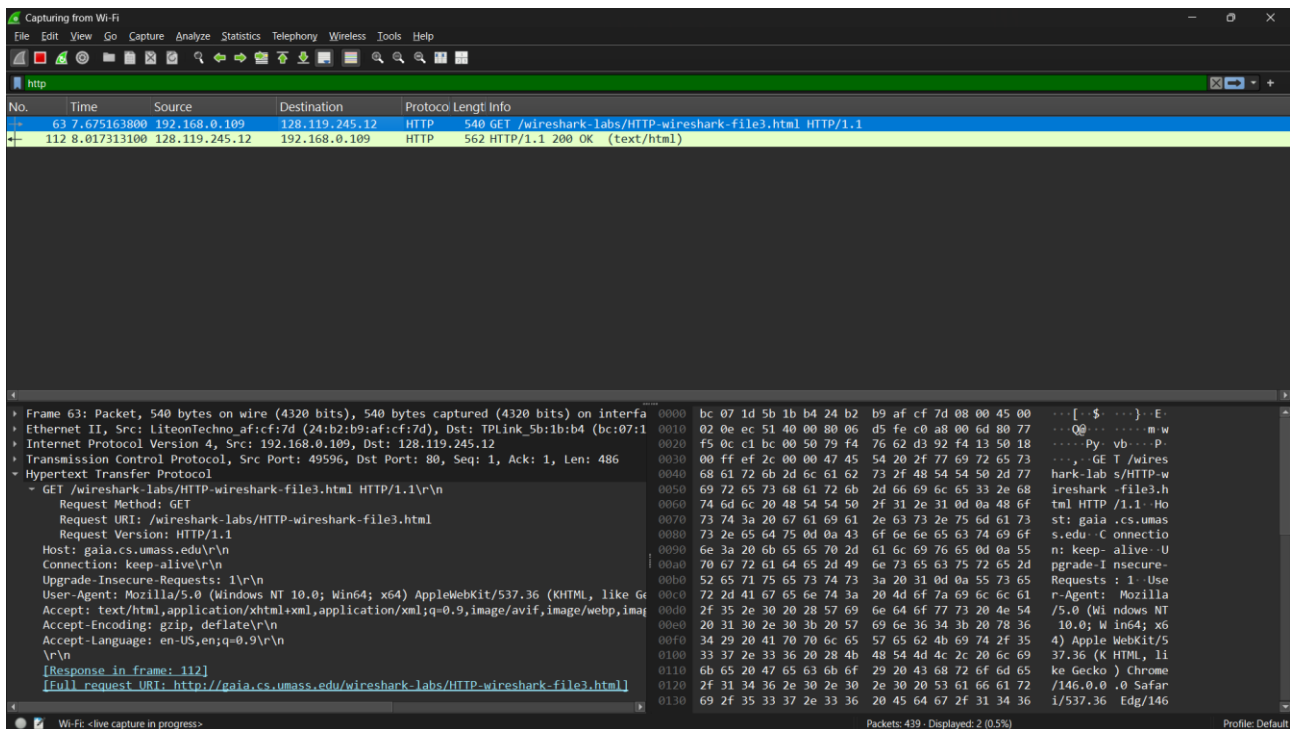
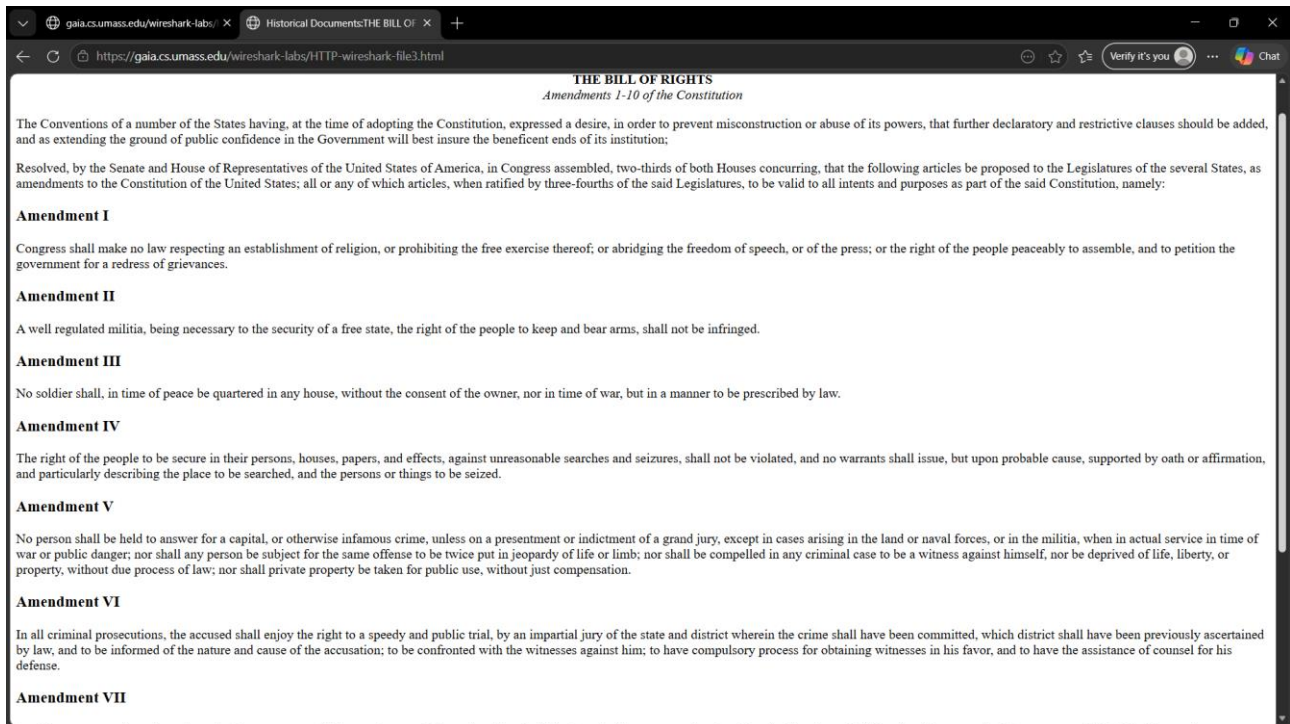
Basic HTTP GET/response interaction



HTTP CONDITIONAL GET/response interaction



Retrieving Long Documents



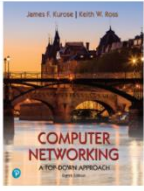
HTML Documents dengan Embedded Objects

Lab2-4 file: Embedded URLs

https://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html

 Pearson

This little HTML file is being served by gaia.cs.umass.edu. It contains two embedded images. The image above, also served from the gaia.cs.umass.edu web site, is the logo of our publisher, Pearson. The image of our 8th edition book cover below is stored at, and served from, a WWW server kurose.cslash.net in France:



And while we have your attention, you might want to take time to check out the available open resources for this book at http://gaia.cs.umass.edu/kurose_ross.

*Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
69	3.551749600	192.168.0.109	128.119.245.12	HTTP	540	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
92	4.010879900	128.119.245.12	192.168.0.109	HTTP	1358	HTTP/1.1 200 OK (text/html)
93	4.023403400	192.168.0.109	128.119.245.12	HTTP	486	GET /pearson.png HTTP/1.1
98	4.586622800	128.119.245.12	192.168.0.109	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
118	5.039168300	192.168.0.109	2.56.99.24	HTTP	453	GET /8E_cover_small.jpg HTTP/1.1
526	6.533890100	2.56.99.24	192.168.0.109	HTTP	688	HTTP/1.1 200 OK (JPEG JFIF image)

Frame 69: Packet, 540 bytes on wire (4320 bits), 540 bytes captured (4320 bits) on interface 0

Ethernet II, Src: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink_5b:1b:b4 (bc:07:12:14:5b:1b:b4)

Internet Protocol Version 4, Src: 192.168.0.109, Dst: 128.119.245.12

Transmission Control Protocol, Src Port: 62513, Dst Port: 80, Seq: 1, Ack: 1, Len: 486

Hypertext Transfer Protocol

GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1\r\n

Request Method: GET

Request URI: /wireshark-labs/HTTP-wireshark-file4.html

Request Version: HTTP/1.1

Host: gaia.cs.umass.edu\r\n

Connection: keep-alive\r\n

Upgrade-Insecure-Requests: 1\r\n

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/106.0.0.0 Safari/537.36

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/svg+xml,*/*;q=0.8\r\n

Accept-Encoding: gzip, deflate\r\n

Accept-Language: en-US,en;q=0.9\r\n

\r\n

[Response in frame: 92]

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html]

Packets: 1585 - Displayed: 6 (0.4%) - Dropped: 0 (0.0%)

Profile: Default

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
69	3.551749600	192.168.0.109	128.119.245.12	HTTP	540	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
92	4.010879900	128.119.245.12	192.168.0.109	HTTP	1358	HTTP/1.1 200 OK (text/html)
93	4.023403400	192.168.0.109	128.119.245.12	HTTP	486	GET /pearson.png HTTP/1.1
98	4.586622800	128.119.245.12	192.168.0.109	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
118	5.039168300	192.168.0.109	2.56.99.24	HTTP	453	GET /8E_cover_small.jpg HTTP/1.1
526	6.533890100	2.56.99.24	192.168.0.109	HTTP	688	HTTP/1.1 200 OK (JPEG JFIF image)

Frame 93: Packet, 486 bytes on wire (3888 bits), 486 bytes captured (3888 bits) on interface 0

Ethernet II, Src: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink_5b:1b:b4 (bc:07:11:00:00:00)

Internet Protocol Version 4, Src: 192.168.0.109, Dst: 128.119.245.12

Transmission Control Protocol, Src Port: 62513, Dst Port: 80, Seq: 487, Ack: 1305, Len: 43

Hypertext Transfer Protocol

- GET /pearson.png HTTP/1.1\r\n
 - Request Method: GET
 - Request URI: /pearson.png
 - Request Version: HTTP/1.1
 - Host: gaia.cs.umass.edu\r\n
 - Connection: keep-alive\r\n
 - User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/14.0.0.0 Safari/537.36 Edg/14.0.0.0\r\n
 - Accept: image/avif,image/webp,image/apng,image/svg+xml,image/*,*/*;q=0.8\r\n
 - Referer: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html\r\n
 - Accept-Encoding: gzip, deflate\r\n
 - Accept-Language: en-US,en;q=0.9\r\n
 - \r\n

[Response in frame: 98]

[Full request URI: http://gaia.cs.umass.edu/pearson.png]

wireshark_Wi-FiWESM3.pcapng

Packets: 1585 - Displayed: 6 (0.4%) - Dropped: 0 (0.0%)

Profile: Default

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
69	3.551749600	192.168.0.109	128.119.245.12	HTTP	540	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
92	4.010879900	128.119.245.12	192.168.0.109	HTTP	1358	HTTP/1.1 200 OK (text/html)
93	4.023403400	192.168.0.109	128.119.245.12	HTTP	486	GET /pearson.png HTTP/1.1
98	4.586622800	128.119.245.12	192.168.0.109	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
118	5.039168300	192.168.0.109	2.56.99.24	HTTP	453	GET /8E_cover_small.jpg HTTP/1.1
526	6.533890100	2.56.99.24	192.168.0.109	HTTP	688	HTTP/1.1 200 OK (JPEG JFIF image)

Frame 93: Packet, 486 bytes on wire (3888 bits), 486 bytes captured (3888 bits) on interface 0

Ethernet II, Src: LiteonTechno_af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink_5b:1b:b4 (bc:07:11:00:00:00)

Internet Protocol Version 4, Src: 192.168.0.109, Dst: 128.119.245.12

Transmission Control Protocol, Src Port: 62513, Dst Port: 80, Seq: 487, Ack: 1305, Len: 43

Hypertext Transfer Protocol

- GET /pearson.png HTTP/1.1\r\n
 - Request Method: GET
 - Request URI: /pearson.png
 - Request Version: HTTP/1.1
 - Host: gaia.cs.umass.edu\r\n
 - Connection: keep-alive\r\n
 - User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/14.0.0.0 Safari/537.36 Edg/14.0.0.0\r\n
 - Accept: image/avif,image/webp,image/apng,image/svg+xml,image/*,*/*;q=0.8\r\n
 - Referer: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html\r\n
 - Accept-Encoding: gzip, deflate\r\n
 - Accept-Language: en-US,en;q=0.9\r\n
 - \r\n

[Response in frame: 98]

[Full request URI: http://gaia.cs.umass.edu/pearson.png]

wireshark_Wi-FiWESM3.pcapng

Packets: 1585 - Displayed: 6 (0.4%) - Dropped: 0 (0.0%)

Profile: Default

Wi-Fi capture window showing HTTP traffic. The packet list table includes columns for No., Time, Source, Destination, Protocol, Length, and Info. Packet 118 is selected, showing details for a GET request to /8E_cover_small.jpg. The packet details pane shows the full request structure, including headers like Host, User-Agent, and Accept.

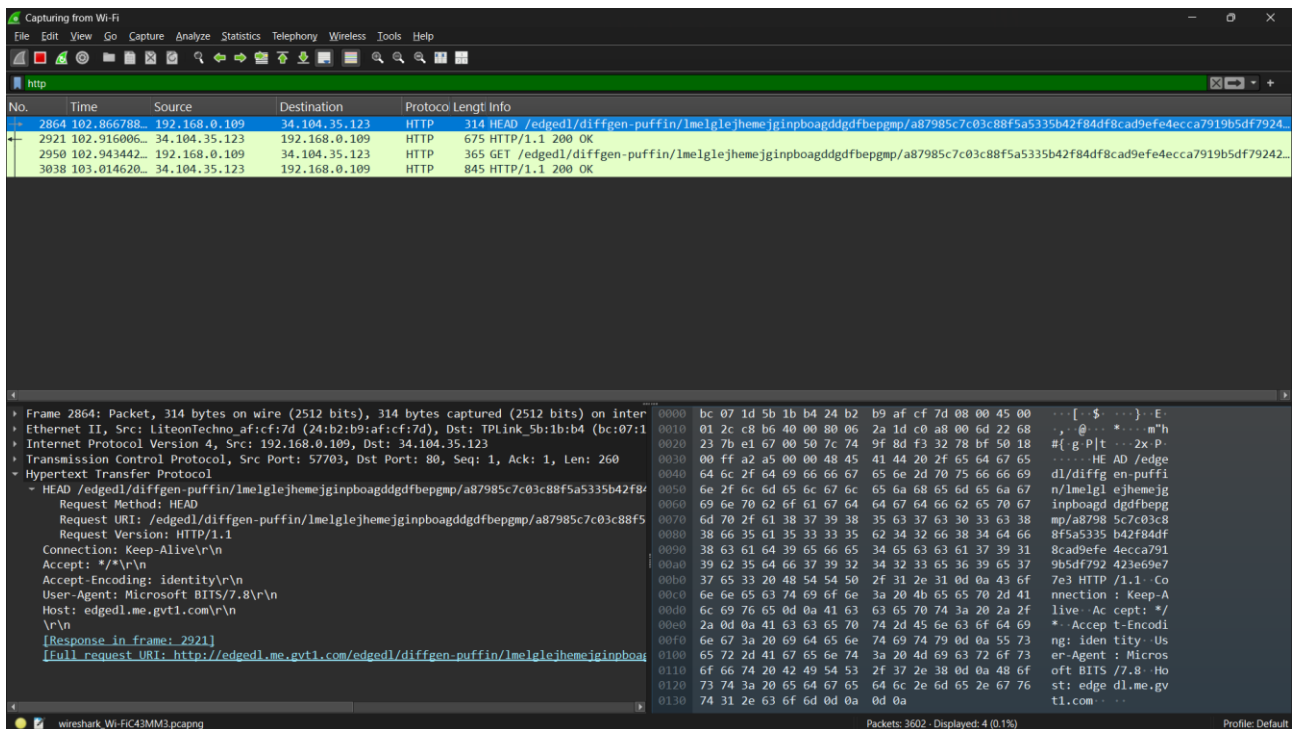
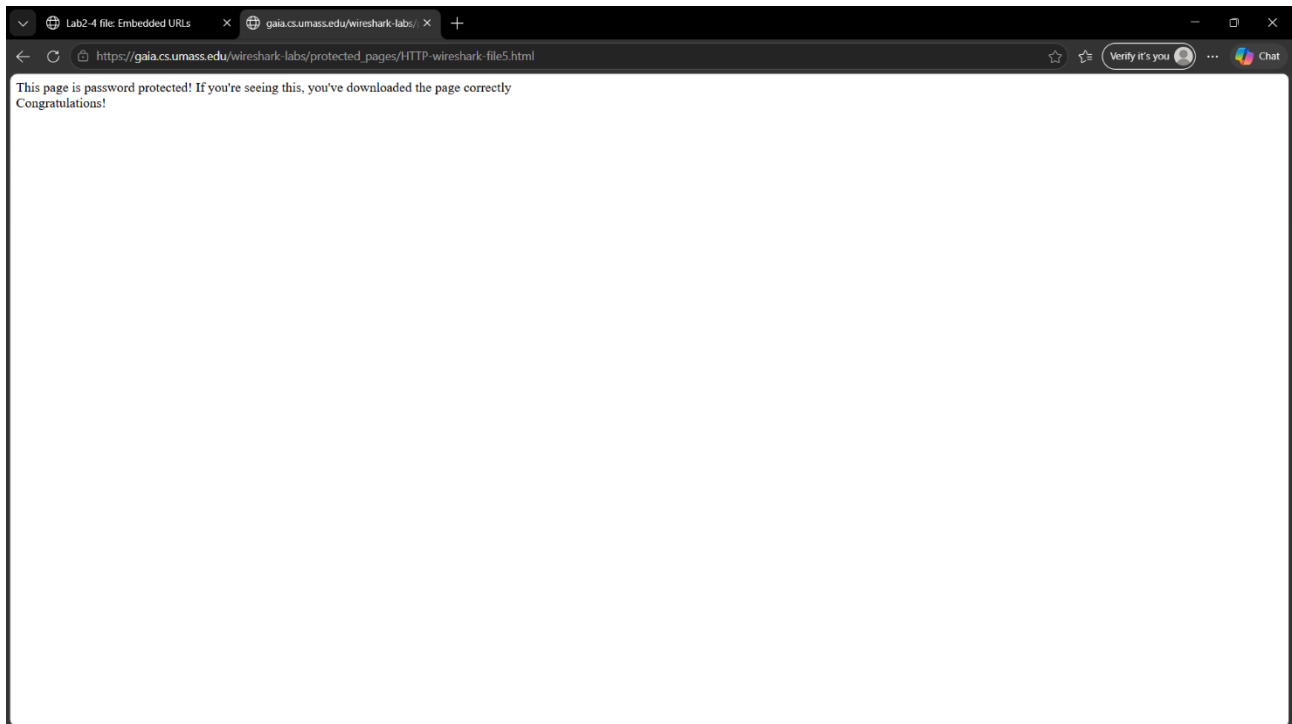
No.	Time	Source	Destination	Protocol	Length	Info
69	3.551749600	192.168.0.109	128.119.245.12	HTTP	540	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
92	4.010879900	128.119.245.12	192.168.0.109	HTTP	1358	HTTP/1.1 200 OK (text/html)
93	4.023403400	192.168.0.109	128.119.245.12	HTTP	486	GET /pearson.png HTTP/1.1
98	4.586622800	128.119.245.12	192.168.0.109	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)
118	5.039168300	192.168.0.109	2.56.99.24	HTTP	453	GET /8E_cover_small.jpg HTTP/1.1
526	6.533890100	2.56.99.24	192.168.0.109	HTTP	688	HTTP/1.1 200 OK (JPEG JFIF image)

Wi-Fi capture window showing the same HTTP traffic, but with packet 526 selected. The packet details pane displays the JPEG File Interchange Format structure, including markers for the start of the image and various segments like quantization tables and Huffman tables.

JPEG File Interchange Format

- Marker: Start of Image (0xFFD8)
- Marker segment: Reserved for application segments - 1 (0xFFE1)
- Marker segment: Reserved for application segments - 13 (0xFFFD)
- Marker segment: Reserved for application segments - 1 (0xFFE1)
- Marker segment: Reserved for application segments - 2 (0xFFE2)
- Marker segment: Reserved for application segments - 14 (0xFFEE)
- Marker segment: Define quantization table(s) (0xFFD9)
- Start of Frame header: Start of Frame (non-differential, Huffman coding) - Baseline DC
- Marker segment: Define restart interval (0xFFD0)
- Marker segment: Define Huffman table(s) (0xFFC4)
- Start of Segment header: Start of Scan (0xFFDA)
- Entropy-coded segment (dissection is not yet implemented) [...] f59b170cff0019ea7b9cf41
- Marker: Restart interval termination - Restart with modulo 8 count 0 (0xFFD0)
- Entropy-coded segment (dissection is not yet implemented) [...] f5bb2f0c6ff00339dfdbcd70
- Marker: Restart interval termination - Restart with modulo 8 count 1 (0xFFD1)
- Entropy-coded segment (dissection is not yet implemented) [...] f5e30f8d8e673bdb79c2d71
- Marker: Restart interval termination - Restart with modulo 8 count 2 (0xFFD2)
- Entropy-coded segment (dissection is not yet implemented) [...] f613afc6df339dcbcf95bc7
- Marker: Restart interval termination - Restart with modulo 8 count 3 (0xFFD3)

HTTP Authentication



Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
2864	102.866788...	192.168.0.109	34.104.35.123	HTTP	314	HEAD /edged1/diffgen-puffin/lme1glejhemegjinpboagddgdfbepgmp/a87985c7c03c88f5a5335b42f84df8cad9efe4ecca7919b5df7924...
2921	102.916006...	34.104.35.123	192.168.0.109	HTTP	675	HTTP/1.1 200 OK
2950	102.943442...	192.168.0.109	34.104.35.123	HTTP	365	GET /edged1/diffgen-puffin/lme1glejhemegjinpboagddgdfbepgmp/a87985c7c03c88f5a5335b42f84df8cad9efe4ecca7919b5df7924...
3038	103.014620...	34.104.35.123	192.168.0.109	HTTP	845	HTTP/1.1 200 OK

content-length: 42278\r\n
[Content length: 42278]
content-security-policy: default-src 'none'\r\n
server: Google-Edge-Cache\r\n
x-content-type-options: nosniff\r\n
x-frame-options: SAMEORIGIN\r\n
x-identity-content-length: 42278\r\n
x-xss-protection: 0\r\n
x-request-id: 9ad163f2-2516-427c-8eac-aadaf615c454\r\n
date: Thu, 02 Apr 2026 14:58:54 GMT\r\n
age: 47287\r\n
last-modified: Thu, 02 Apr 2026 14:55:43 GMT\r\n
etag: "5b006d1"\r\n
content-type: application/octet-stream\r\n
vary: Accept-Encoding\r\n
alt-svc: h3=":443"; ma=2592000, h3-29=":443"; ma=2592000\r\n
cache-control: no-transform,public,max-age=86400\r\n
\r\n
[Request in frame: 2864]
[Time since request: 49.319300 milliseconds]

0000 24 b2 b9 af cf 7d bc 07 1d 5b 1b b4 08 00 45 00 \$....}...[...E
0010 02 95 87 72 00 00 3b 06 ee f8 22 68 23 7b c0 a8 ...r;..."h#
0020 00 6d 00 50 e1 67 f3 32 78 bf 7c 74 a0 91 50 18 ...mPg2x|tP
0030 04 1a 16 87 00 00 48 54 54 50 2f 31 2e 31 20 32HT TP/1.1 2
0040 30 30 20 4f 4b 0d 0a 61 63 63 65 70 74 2d 72 61 00 OK;a ccept-ra
0050 6e 67 65 73 3a 20 62 79 74 65 73 0d 0a 63 6f 6e nges: by tes:con
0060 74 65 6e 74 2d 64 69 73 70 6f 73 69 74 69 6f 6e tent-dis position
0070 3a 20 61 74 74 61 63 68 6d 65 6e 74 0d 0a 63 6f : attach ment-co
0080 6e 74 65 6e 74 2d 6c 65 6e 67 74 68 3a 20 34 32 ntent-le ngth: 42
0090 32 37 38 0d 0a 63 6f 6e 74 65 6e 74 2d 73 65 63 278 con tent-sec
00a0 75 72 69 74 79 2d 70 6f 6e 69 63 79 3a 20 64 65 urity-po licy: de
00b0 66 61 75 6e 74 2d 73 72 63 20 27 6e 6f 6e 65 27 fault-sr c 'none'
00c0 0d 0a 73 65 72 76 65 72 3a 20 47 6f 6f 67 6c 65 ..server : Google
00d0 2d 45 64 67 65 2d 43 61 63 68 65 0d 0a 78 2d 63 -Edge-Ca che x-c
00e0 6f 6e 74 65 6e 74 2d 74 79 70 65 2d 6f 70 74 69 ontent-t ype-opti
00f0 6f 6e 73 3a 20 6e 6f 73 6e 69 66 66 0d 0a 78 2d ons: nos niff x-
0100 66 72 61 6d 65 2d 6f 70 74 69 6f 6e 73 3a 20 53 frame-op tions: S
0110 41 4d 45 4f 52 49 47 49 4e 0d 0a 78 2d 69 64 65 AMEORIGI N x-ide
0120 6e 74 69 74 79 2d 63 6f 6e 74 65 6e 74 2d 6c 65 ntity-co ntent-le
0130 6e 67 74 68 3a 20 34 32 37 38 0d 0a 78 2d 78 ngth: 42 278 x-x

wireshark_Wi-Fi43MM3.pcapng

Packets: 4776 - Displayed: 4 (0.1%) - Dropped: 0 (0.0%)

Profile: Default